

Digital Signal Processing (DSP) - Session 2

I. Introduction

1. The Concept of Frequency in Continuous-Time and Discrete-Time Signals

1.1. Continuous-Time Sinusoidal Signals A simple harmonic oscillation is mathematically described by a continuous-time sinusoidal signal. A continuous-time sinusoid is completely characterized by three parameters: amplitude (A), frequency (Ω or F), and phase (θ).

General Expression (Radians):

$$x(t) = A \cos(\Omega t + \theta), \quad -\infty < t < \infty$$

where

- A is the amplitude.
- Ω is the angular frequency in **radians per second (rad/s)**.
- t is time in **seconds**.
- θ is the phase in **radians**.

Frequency Relation:

$$\Omega = 2\pi F$$

where

- F is the frequency in **cycles per second (Hz)**.

General Expression (Hertz):

$$x(t) = A \cos(2\pi Ft + \theta), \quad -\infty < t < \infty$$

Periodicity: For every fixed value of F , the signal is periodic.

$$x(t + T) = x(t)$$

The fundamental period is $T = 1/F$.

Uniqueness: Continuous-time sinusoidal signals with distinct frequencies are distinct.

Rate of Oscillation: Increasing F increases the rate of oscillation (more cycles within a given time interval). Since time is continuous, the frequency F can increase without bound.

Complex Exponential Form and Phasors:

$$x(t) = Ae^{j(\Omega t + \theta)}$$

Sinusoidal signals are closely related to complex exponential signals through **Euler's identity**:

$$e^{j\phi} = \cos \phi + j \sin \phi$$

A real sinusoidal signal can be expressed as the sum of two equal-amplitude complex-conjugate exponentials:

$$x(t) = A \cos(\Omega t + \theta) = \frac{A}{2} e^{j(\Omega t + \theta)} + \frac{A}{2} e^{-j(\Omega t + \theta)}$$

Frequency Range: Although physical frequency is nonnegative, both positive and negative frequencies are used for mathematical convenience.

$$-\infty < F < \infty$$

1.2. Discrete-Time Sinusoidal Signals A discrete-time sinusoidal signal is a sequence of numbers derived from a function of an integer variable n .

General Expression (Radians):

$$x[n] = A \cos(\omega n + \theta), \quad -\infty < n < \infty$$

where

- A is the amplitude.
- ω is the angular frequency in **radians per sample**.
- n is the **sample index** (an integer).
- θ is the phase in **radians**.

Frequency Relation:

$$\omega = 2\pi f$$

where

- f is the frequency in **cycles per sample**.

General Expression (Cycles):

$$x[n] = A \cos(2\pi f n + \theta), \quad -\infty < n < \infty$$

Periodicity: A discrete-time sinusoid is periodic **only if** its normalized frequency f is a **rational number**. A signal is periodic with period N ($N > 0$) if

$$x[n + N] = x[n]$$

This requires $f_0 = k/N$ where k and N are integers. The fundamental period is obtained by expressing f_0 as a reduced fraction k/N , where k and N are relatively prime.

Uniqueness and Aliasing: Discrete-time sinusoids whose frequencies differ by an integer multiple of 2π are identical.

$$\cos[(\omega_0 + 2\pi)n + \theta] = \cos(\omega_0 n + 2\pi n + \theta) = \cos(\omega_0 n + \theta)$$

Therefore, every frequency outside the principal interval is an **alias** of a frequency inside it.

The unique frequency ranges are commonly chosen as $-\pi \leq \omega \leq \pi$ or equivalently $-\frac{1}{2} \leq f \leq \frac{1}{2}$.

Highest Rate of Oscillation: The highest rate of oscillation occurs at $\omega = \pi$ or $\omega = -\pi$. This corresponds to $f = \frac{1}{2}$ or $f = -\frac{1}{2}$.

Observations:

- Increasing ω from 0 to π increases the rate of oscillation.
- Increasing ω from π to 2π decreases the apparent rate of oscillation because of aliasing.

Complex Exponential Form: Using Euler's identity, a discrete-time sinusoid can be expressed as the sum of two complex-conjugate exponentials:

$$x[n] = A \cos(\omega n + \theta) = \frac{A}{2} e^{j(\omega n + \theta)} + \frac{A}{2} e^{-j(\omega n + \theta)}$$

Frequency Range: Unlike continuous-time sinusoids, discrete-time frequencies repeat every 2π . The frequency axis is periodic with period 2π .

Two commonly used principal frequency intervals are $0 \leq \omega \leq 2\pi$ and $-\pi \leq \omega \leq \pi$.

2. Analog-to-Digital Conversion (ADC)

2.1. Basic parts

- Goal: convert a continuous-time, continuous-amplitude signal into a discrete-time, discrete-amplitude sequence.
- Typical chain: **anti-aliasing filter** → **sampling** → **quantization** → **encoding**.

1. Sampling

- Sample at rate f_s samples/second (sampling period $T_s = 1/f_s$):

$$x[n] = x_a(nT_s)$$

- Sampling a signal multiplies it by an impulse train in time, which creates **spectral replicas** in frequency.
- In discrete-time, the corresponding digital frequency is $\omega = 2\pi f/f_s$.

2. Quantization

- Rounds each sample to the nearest value on a finite set of amplitude levels.
- Quantization step size (uniform quantizer):

$$\Delta = \frac{V_{\max} - V_{\min}}{2^B}$$

where B is number of bits.

- Quantization error:

$$e_q[n] = x[n] - x_q[n].$$

3. Encoding

- Maps each quantized level to a binary word of length B .
- Bit rate (for one channel): $R = Bf_s$ bits/second.

2.2. Nyquist–Shannon sampling theorem & Aliasing

- If a continuous-time signal is bandlimited to $|f| \leq B$ Hz, then perfect reconstruction is possible when $f_s \geq 2B$ (Nyquist rate).
- If $f_s < 2B$, spectral replicas overlap $\rightarrow$ **aliasing**, meaning high-frequency content folds into lower frequencies.
- Practical note: because real signals are not perfectly bandlimited, we usually use an **anti-aliasing low-pass filter** before sampling.

2.3. Signal-to-quantization noise ratio (SQNR)

- SQNR measures the ratio of signal power to quantization-noise power:

$$\text{SQNR} = \frac{P_{\text{signal}}}{P_{\text{quant}}} = \frac{3}{2} \cdot 2^{2B}$$

- SQNR is also often expressed in decibels (dB):

$$\text{SQNR}_{\text{dB}} = 10 \log_{10}(\text{SQNR}) \approx 1.76 + 6.02B$$

Tip (logarithmic scale / dB): Use a log scale when quantities span many orders of magnitude (very large or very small).

- **Power ratio (dB):** $P_{\text{dB}} = 10 \log_{10}\left(\frac{P_2}{P_1}\right)$
- **Amplitude ratio (dB):** $A_{\text{dB}} = 20 \log_{10}\left(\frac{A_2}{A_1}\right)$ (because $P \propto A^2$)

II. Discrete-Time Signals

1. Representations

Discrete-time signals $x[n]$ are defined only for integer values of the independent variable $n \in \mathbb{Z}$. They can be represented in four standard forms:

- **Graphical:** A stem plot displaying sample values (amplitudes) as discrete vertical lines plotted against integer indices n .
- **Functional:** A piecewise or closed-form mathematical expression defining $x[n]$ for all n .
– Example: $x[n] = 2^n u[n]$
- **Tabular:** A table listing discrete sample values directly corresponding to specific values of n .

- **Sequence (Vector):** An ordered set of numbers enclosed in brackets, where an arrow or bold term indicates the sample index $n = 0$.
 - Example: $x[n] = \{\dots, 0, \underset{\uparrow}{2}, 1, 3, 0, \dots\}$ (the arrow highlights $x[0] = 2$).

2. Some Elementary Discrete-Time Signals

2.1. Unit Sample Signal (Impulse Signal) Defines a single isolated spike of unit amplitude at $n = 0$:

$$\delta[n] = \begin{cases} 1, & \text{if } n = 0 \\ 0, & \text{if } n \neq 0 \end{cases}$$

2.2. Unit Step Signal Represents a signal that gets activated at $n = 0$ with constant unit amplitude:

$$u[n] = \begin{cases} 1, & \text{if } n \geq 0 \\ 0, & \text{if } n < 0 \end{cases}$$

- *Relationship to Impulse:*

$$u[n] = \sum_{k=-\infty}^n \delta[k] = \sum_{k=0}^{\infty} \delta[n - k]$$

2.3. Unit Ramp Signal A signal that grows linearly with sample index n for non-negative time:

$$r[n] = \begin{cases} n, & \text{if } n \geq 0 \\ 0, & \text{if } n < 0 \end{cases}$$

- *Relationship to Unit Step:*

$$r[n] = n \cdot u[n]$$

2.4. Exponential Signals General discrete-time exponential signals take the form $x[n] = a^n$:

- **Real Exponential** ($x[n] = a^n$, where $a \in \mathbb{R}$):
 - $|a| > 1$: Growing exponential.
 - $|a| < 1$: Decaying exponential.
 - $a < 0$: Alternating signs between positive and negative samples.
- **Complex Exponential** ($x[n] = e^{j\omega_0 n}$):
 - Expressed via Euler's formula: $e^{j\omega_0 n} = \cos(\omega_0 n) + j \sin(\omega_0 n)$.
 - Unlike continuous complex exponentials, discrete complex exponentials are **periodic only if** $\frac{\omega_0}{2\pi}$ is a rational number.

Example: Sketch the graph of this signal:

$$x[n] = u[n - 2] - u[n - 5]$$

```
import numpy as np
import matplotlib.pyplot as plt

# Define the index range
n = np.arange(-2, 9)

# Define step function u[n]
def u(n):
    return np.where(n >= 0, 1, 0)

# Signals
u_n2 = u(n - 2)
u_n5 = u(n - 5)
x_n = u_n2 - u_n5

# Plot step-by-step
fig, axes = plt.subplots(3, 1, figsize=(8, 7), sharex=True)

# Plot 1: u[n-2]
axes[0].stem(n, u_n2, basefmt="C0-")
axes[0].set_ylabel('$u[n-2]$', fontsize=12)
axes[0].set_title('Step 1: First Shifted Unit Step $u[n-2]$', fontsize=12)
axes[0].set_yticks([0, 1])
axes[0].grid(True, linestyle='--', alpha=0.6)

# Plot 2: u[n-5]
axes[1].stem(n, u_n5, basefmt="C0-")
axes[1].set_ylabel('$u[n-5]$', fontsize=12)
axes[1].set_title('Step 2: Second Shifted Unit Step $u[n-5]$', fontsize=12)
axes[1].set_yticks([0, 1])
axes[1].grid(True, linestyle='--', alpha=0.6)

# Plot 3: x[n] = u[n-2] - u[n-5]
axes[2].stem(n, x_n, basefmt="C0-")
axes[2].set_ylabel('$x[n]$', fontsize=12)
axes[2].set_xlabel('Sample Index $n$', fontsize=12)
axes[2].set_title('Step 3: Resulting Signal $x[n] = u[n-2] - u[n-5]$', fontsize=12)
axes[2].set_yticks([0, 1])
axes[2].grid(True, linestyle='--', alpha=0.6)

plt.tight_layout()
plt.savefig("signal_visualization.png", dpi=300)
plt.show()
```

```
print("Signal values x[n]:")
for ni, val in zip(n, x_n):
    print(f"n={ni:2d} : {val}")
```

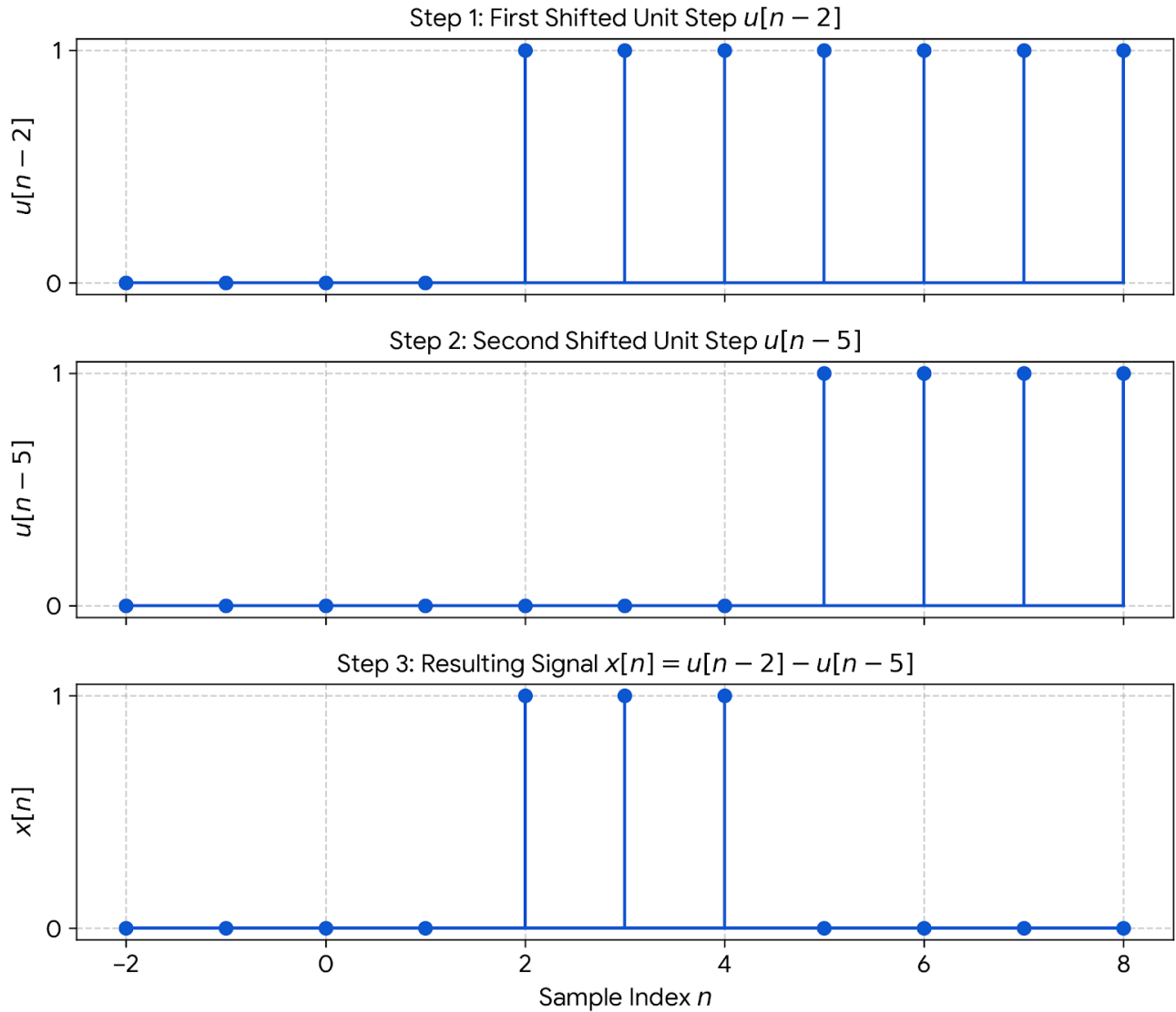


Figure 1: signal_visualization.png

3. Classification of discrete-time signals

3.1. Energy signals and power signals

- Total Energy:

$$E = \sum_{n=-\infty}^{\infty} |x[n]|^2$$

- **Average Power:**

$$P = \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=-N}^N |x[n]|^2$$

- An **energy signal** has a **finite, non-zero total energy** ($0 < E < \infty$), which forces its average power to be **zero** ($P = 0$).
- A **power signal** has a **finite, non-zero average power** ($0 < P < \infty$), which means its total energy is **infinite** ($E = \infty$).
- Signals that grow infinitely over time are neither energy nor power signals ($E = \infty, P = \infty$).

Example:

- Unit Sample Signal $\delta[n]$:
 - **Total Energy (E):**

$$E = \sum_{n=-\infty}^{\infty} |\delta[n]|^2 = |\delta[0]|^2 = 1^2 = 1$$

- **Average Power (P):**

$$P = \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=-N}^N |\delta[n]|^2 = \lim_{N \rightarrow \infty} \frac{1}{2N+1} (1) = 0$$

Since $0 < E < \infty$ and $P = 0$, $\delta[n]$ is an **energy signal**.

- Unit Step Signal $u[n]$:
 - **Total Energy (E):**

$$E = \sum_{n=-\infty}^{\infty} |u[n]|^2 = \sum_{n=0}^{\infty} 1^2 = 1 + 1 + 1 + \dots = +\infty$$

Because energy is infinite, $u[n]$ cannot be an energy signal.

- **Average Power (P):**

In the interval $[-N, N]$, the signal is 1 only for $n = 0, 1, 2, \dots, N$ (which contains $N+1$ active samples):

$$P = \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=0}^N 1 = \lim_{N \rightarrow \infty} \frac{N+1}{2N+1}$$

Dividing numerator and denominator by N :

$$P = \lim_{N \rightarrow \infty} \frac{1 + 1/N}{2 + 1/N} = \frac{1 + 0}{2 + 0} = \frac{1}{2}$$

Since $0 < P < \infty$, $u[n]$ is a **power signal**.

3.2. Periodic signals and aperiodic signals

- A signal $x[n]$ is periodic if there exists a positive integer fundamental period $N \in \mathbb{Z}^+$ such that $x[n+N] = x[n]$ for all integer indices n .

- The smallest positive integer N satisfying this condition is called the **fundamental period**.
- If no such integer N exists, the signal is aperiodic.
- All periodic signals with non-zero values over an infinite duration are **power signals**.
- For discrete complex exponentials $x[n] = e^{j\omega_0 n}$ or sinusoids $x[n] = \cos(\omega_0 n)$, periodicity requires that $\frac{\omega_0}{2\pi} = \frac{k}{N}$ (a rational ratio), where the fundamental period is $N = \frac{2\pi k}{\omega_0}$.

3.3. Symmetric (even) and antisymmetric (odd) signals

- A signal is **even (symmetric)** if $x[-n] = x[n]$ for all n (reflection across the vertical axis $n = 0$).
- A signal is **odd (antisymmetric)** if $x[-n] = -x[n]$ for all n (forces $x[0] = 0$).
- **Decomposition:** Any arbitrary discrete-time signal $x[n]$ can be uniquely split into a sum of its even part $x_e[n]$ and odd part $x_o[n]$:

$$x[n] = x_e[n] + x_o[n]$$

where:

$$x_e[n] = \frac{1}{2}(x[n] + x[-n])$$

$$x_o[n] = \frac{1}{2}(x[n] - x[-n])$$

4. Simple Manipulations of Discrete-Time Signals

4.1. Time Shifting & Time Reversal

$$x[\pm n \pm a], a \in \mathbb{N}$$

- **Time Shifting** ($x[n \pm a]$): Moving the signal along the time axis without changing its shape.
 - $x[n - a]$ delays (shifts right) the signal by a units.
 - $x[n + a]$ advances (shifts left) the signal by a units.
- **Time Reversal / Folding** ($x[-n]$): Reflects the signal about $n = 0$.
- **Combined Shift & Reversal** ($x[-n \pm a]$): Order of operations matters:
 - First shift $x[n]$ to get $x[n \pm a]$, then reverse time $n \rightarrow -n$ to get $x[-n \pm a]$.
 - Alternatively, factor as $x[-(n \mp a)]$: first fold $x[n] \rightarrow x[-n]$, then shift in the direction dictated by the factored sign.

4.2. Up-sampling & Down-sampling

$$x[cn], c \in \mathbb{N}$$

- **Down-sampling / Decimation** ($x[cn], c \in \mathbb{N}, c > 1$):
 - Keeps every c -th sample and discards the rest ($c - 1$ samples dropped between kept samples).
 - Shrinks/compresses the signal in time domain and leads to spectrum expansion in frequency domain (potential aliasing if not pre-filtered).
- **Up-sampling / Interpolation** ($c \in \mathbb{N}, c > 1$):

- Expressed as $x_u[n] = \begin{cases} x[n/c], & \text{if } n \text{ is a multiple of } c \\ 0, & \text{otherwise} \end{cases}$
- Inserts $c - 1$ zero-valued samples between consecutive original samples, expanding the signal in the time domain.

III. Discrete-Time Systems

1. Representations

1.1. Input-Output Description of Systems

- A discrete-time system operates on an input sequence $x[n]$ to produce an output sequence $y[n]$, represented as $y[n] = \mathcal{T}\{x[n]\}$.
- Systems are represented by difference equations: Linear constant-coefficient difference equations (LCCDE) relate the current output to past outputs and current/past inputs:

$$\sum_{k=0}^N a_k y[n-k] = \sum_{m=0}^M b_m x[n-m]$$

1.2. Block Diagram Representation of Discrete-Time Systems

- Constructed using three basic functional building blocks:
 - **Adder:** Adds two signals, $y[n] = x_1[n] + x_2[n]$.
 - **Constant Multiplier (Gain):** Scales a signal by a constant, $y[n] = a \cdot x[n]$.
 - **Unit Delay Element (z^{-1}):** Delays a signal by one sample, $y[n] = x[n-1]$. (A unit advance z yields $x[n+1]$).

2. Classification of Discrete-Time Systems

2.1. Static vs dynamic systems

- **Static (Memoryless):** The output $y[n]$ at index n depends *only* on the input $x[n]$ at that same index n (e.g., $y[n] = Cx^2[n]$).
- **Dynamic (With Memory):** The output $y[n]$ depends on past or future values of the input or output (e.g., $y[n] = x[n] + x[n-1]$).

2.2. Time-invariant vs time-variant systems

- **Time-Invariant (Shift-Invariant):** A time shift in the input causes an identical time shift in the output: if $\mathcal{T}\{x[n]\} = y[n]$, then $\mathcal{T}\{x[n-k]\} = y[n-k]$ for all integer k .
- **Time-Variant:** The system's input-output relationship changes over time (e.g., coefficients depend on n , as in $y[n] = n \cdot x[n]$, or time operations occur, as in $y[n] = x[-n]$).

2.3. Linear vs nonlinear systems

- **Linear:** Satisfies the principle of **superposition** (both additivity and homogeneity):

$$\mathcal{T}\{a \cdot x_1[n] + b \cdot x_2[n]\} = a \cdot \mathcal{T}\{x_1[n]\} + b \cdot \mathcal{T}\{x_2[n]\}$$

- **Nonlinear:** Fails superposition (e.g., contains powers, square roots, absolute values, or non-zero initial conditions like $y[n] = x^2[n]$ or $y[n] = x[n] + 3$).

2.4. Causal vs non-causal systems

- **Causal:** The output $y[n]$ at any index n depends *only* on present and/or past inputs $x[k]$ ($k \leq n$). It cannot depend on future inputs.
- **Non-Causal:** The output $y[n]$ depends on future input samples $x[k]$ ($k > n$) (e.g., $y[n] = x[n + 1]$). Non-causal systems cannot be implemented in real time, but can process pre-recorded data.

2.5. Stable vs unstable systems

- **BIBO Stable (Bounded-Input, Bounded-Output):** Every bounded input $|x[n]| \leq M_x < \infty$ produces a bounded output $|y[n]| \leq M_y < \infty$.
- **LTI System Stability Condition:** An LTI system with impulse response $h[n]$ is BIBO stable if and only if its impulse response is **absolutely summable**:

$$\sum_{n=-\infty}^{\infty} |h[n]| < \infty$$

Next session: Analysis of Discrete-Time Linear Time-Invariant Systems